CHEETAH: archived comparison tables

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These are the original Tables 6, 7 and 10, moved out of the main paper. Their original table numbers and contents are preserved below. They use historical host-backed reference models and retain their original qualifications; they are not the unchanged paper Baseline 2 results in the updated manuscript. They do not establish measured TPU-v3 speedups or physically qualified BF16 chips. This file is standalone and is not included by `neurips_2026.tex`.

Table 6: Archived ratios over the earlier host-backed TPU-v3-resource reference; baseline and BF16 physical-cost reconciliation remain open. Phase ratios do not add. Dashes denote inapplicable phases.

Workload	Overall	Prefill	Decode
EfficientNet-B1	16.53×	–	–
EfficientNet-B2	16.94×	–	–
EfficientNet-B3	18.25×	–	–
EfficientNet-B4	19.14×	–	–
EfficientNet-B5	19.79×	–	–
EfficientNet-B6	20.38×	–	–
EfficientNet-B7	20.94×	–	–
SD1.5 UNet (50 calls)	40.34×	–	–
Llama-1 13B	122.59×	48.00×	148.94×
Llama-1 65B	68.92×	25.36×	80.76×
Llama-3 70B	91.78×	47.14×	105.93×
DeepSeek-V3	31.71×	19.85×	37.93×

Table 7: Uniform full-inference predictions in seconds for all 32 requests. Preparation is included in totals. SW keeps TPU-v3 resources fixed.

Model	Phase	Reference	Reference + SW	Custom	Speedup
Llama-1 13B	Prefill	980.380	720.526	20.425	48.00×
	Decode	17,668.410	1,317.840	118.631	148.94×
	Total	18,663.574	2,053.150	152.245	122.59×
Llama-1 65B	Prefill	4,407.777	pending	173.827	25.36×
	Decode	82,145.424	pending	1,017.176	80.76×
	Total	86,627.355	pending	1,256.908	68.92×
Llama-3 70B	Prefill	15,076.868	10,905.168	319.824	47.14×
	Decode	106,905.123	7,982.193	1,009.241	105.93×
	Total	121,981.992	18,887.361	1,329.065	91.78×

Table 10: Corrected custom-chip predictions (seconds); ratios use the retained, uncalibrated TPU-v3 extension.

Phase	TPU-v3 extension	Custom chip	Speedup
Prefill	13,387.193	1,078.914	12.408×
Decode	53,083.324	2,074.999	25.582×
Preparation	131.252	42.973	3.054×
Total	66,601.769	3,196.886	**20.833**×